The Energy Balance of Extreme Snow ablation

William Rudisill[1], Alan Rhoades[1], Megan Mason[2], Gabriel Lewis[2], Andrew Schwartz[2], Marianne Cowherd[4], Benjamin Hatchett[3], Erica Siriila-Woodburn[1], Dan Feldman[1]

Target Journal: JGR Atmospheres

Abstract
Test

1 Introduction

While meltwater from seasonal snowpacks has historically provided ∼70% of runoff in the western USA (Li et al., 2017), these natural reservoirs and our ability to measure, monitor, and predict them are in decline (Hale et al., 2023; Cowherd et al., 2024). From a practical perspective, many operational snowmelt runoff models use a temperature index approach, whereby air temperature (T_{air}) is the primary predictor of melt (Anderson, 1973). While these models have proven effective (after calibrating empirical coefficients for each site or watershed), the physical justification for such models is not clear. Since at least the 1980s the field of snow hydrology appreciates that understanding the climatic behavior of snow necessitates considering the entire energy budget of the snowpack. In particular, net shortwave radiation incident on the snowpack (SW_{net}) is a first-order predictor of rates of melt, and it is well accepted that both small impurities (Deems et al., 2013) and natural grain-growth mechanisms that decrease snow albedo (α) are responsible, in conjunction with rising sun-angles in the spring, for the majority of energy available for snowmelt. Unlike atmospheric radiation in the longwave portion of the spectra (4-100μm), shortwave (0.3-3μm) is mainly insensitive to variations in T_{air} itself. Variations in $SW\downarrow$ in time (and across elevation) are driven primarily by Rayleigh scattering by O_2 and N_2, Mie scattering by natural occurring and human-made aerosols, dust, and cloud droplets, as well as absorption primarily by water vapor, CO_2, and O_3.

At the same time, recent work has highlighted the strong correlation between extreme snow ablation events and so-called "snow-eater heatwaves" — prolonged periods of sustained (day and night), anomalous, and above freezing T_{air} during the snow season (Rhoades et al., 2026). The characteristics of snow-eater heatwaves have already appreciably changed since the 1850s. These events are occurring more frequently, at a higher intensity, over a larger portion of the western US (WUS), and earlier in the water year.

An example of such an early-season, high-intensity, snow-eater heatwave occurred in March of 2026. This event, which started earlier than the typical window of occurrence studied in Rhoades et al. (2026) resulted in widespread and rapid melting at many snow-observing sites and one to two months earlier than average streamflow in many basins. This raises an apparent paradox: if $SW\downarrow$ provides most of the energy for snowmelt, and yet it is largely insensitive to T_{air}, then why are heatwaves so efficient at ablating snow?

Using the March 2026 event as a case study, we ask the following questions:

- What are the dominant snowmelt energy forcings during early-season heatwaves, and how do they scale with increasing T_{air}?

- How do vegetation canopies and atmospheric conditions — including clouds or lack thereof — shape such energetic forcings during heatwave events?
- What are the relative contributions of sublimation and melt to extreme snow ablation during such events?

For some perspective, approximately half of North America's snow water volume resides under tree canopy (Kim et al., 2021). Canopies absorbs shortwave radiation, increases the longwave radiation, and reduces winds relative to open or non-vegetated snow surfaces, however the magnitudes of these changes can vary widely across space and time.

While low clouds have the most substantive effect on the snow energy balance (Rudisill et al., 2025), despite otherwise drier-than-normal conditions, high altitude cirrus clouds can occur during high-pressure ridging events commonly associated with cool-season heatwaves (Rhoades et al. (2026) Figure 1 and Supplemental Figure S5). Cirrus increases longwave and decreases shortwave relative to clear skies (L'Ecuyer et al., 2019), though the radiative impacts to snowpacks are underexplored in the literature, let alone during heatwave events.

To answer these questions, we use the March 2026 event as an object lesson to explore heatwave-snow interactions and place the record heat and associated snow ablation in a climatic and synoptic context. We focus our analysis on the greater Northern Sierra Nevada, which supports runoff into the American, Yuba, and Lake Tahoe/Truckee River basins and use multiple lines of evidence (reanalyses, long-term station records, remote sensing observations, and models) to understand the energetic components of the anomalous observed snow ablation.

The outcomes from this study answer basic science questions about energy available for snowmelt or sublimation in water-resource critical basins in the northern Sierra Nevada, with implications for other regions. The findings can directly inform efforts towards management and resilience against extreme events by providing actionable steps towards improving the monitoring and prediction of snowmelt. Last, improved understanding of vegetation and snow interactions during heatwaves, as well as during other runoff-inducing events such as typical warm-day melt and rain-on-snow, can inform headwater management practices through improved decision support tools (e.g., Heggli et al. 2022; 2026).

2 Background and Methods

2.1 Study area

We focus our analysis on three basins in the immediate vicinity of the the Central Sierra Snow Laboratory (CSSL; 39.32 °N, 120.37 °W, 2100 m.a.s.l) located in Soda Springs, CA (Schwartz et al., 2026). These include the Yuba, North Fork American, and Truckee River Watersheds (the greater Truckee River Watershed is composed of the Lake Tahoe basin and Truckee River basin at the eight-digit hydrologic unit code (HUC8) scale) (Fig. 1a–d). The Yuba and North Fork American rivers support the New Bullards Bar and Folsom reservoirs, respectively. These reservoirs have combined storage capacities approaching 2 million acre feet and hydropower generation potential of 559 megawatts TODO: citation. All three rivers support varied downstream consumptive uses (i.e., urban, industrial and agricultural) and provide critical aquatic and riparian habitats.

2.2 Observational data

We use data from twelve Natural Resources Conservation Service (NRCS) SNOwpack TELemetry (SNOTEL) Network stations located throughout the study region spanning TODO: X to TODO: Y m.a.s.l. SNOTEL is the primary snow observing program in the western USA (Serreze et al., 1999) and routinely measures snow water equivalent (SWE) and air temperature, though the homogeneity of air temperature measurements remains a challenge for the SNOTEL network (Oyler et al., 2015; McAfee et al., 2019). To supplement SNOTEL temperature records, we use data from four US Historical Climate Reference Network (USHCRN) stations and the gridded nClimGrid product throughout the study region. USHCRN stations are high-quality stations subset from the larger GHCN-D dataset Menne et al. (2012). The HCNv2 dataset provides homogenized and quality-assured monthly air temperatures based on USHCRN data that dates as far back as the late 1800s (Menne et al., 2009). While this data does not have the same spatial density and elevation range as SNOTEL, the longer temporal record and high quality control mean that these observations can more confidently be used for evaluation of climate extremes, such as heatwave events (Bercos-Hickey et al., 2022).

In addition to hosting an NRCS SNOTEL site, CSSL provides long-term meteorological and snowpack records in addition to surface energy balance measurements. Campbell Scientific CNR4 four-component radiometers measure incoming and outgoing SW and LW radiation from the atmosphere and snowpack. A Campbell Scientific IRGASON eddy covariance system measures sensible (H_s) and latent (H_l) heat fluxes above the snowpack. Both the IRGASON and CNR4 are hosted on a movable arm (Fig. 1c) that automatically adjusts to maintain a constant distance of approximately one meter above the snow surface as snow accumulates or ablates throughout the season. Eddy covariance fluxes were processed using the Licor EddyPro v7.0.9 software with despiking, double coordinate rotation, and flux quality assignment. TODO: X% of the H_s and H_l fluxes are retained for analysis during the month of March.

The CSSL forest study site hosts an additional meteorological station approximately 150 m North of the main CSSL site. This tower hosts an additional Kipp and Zonen SP Lite2 radiometer that measures below-canopy up and downwelling shortwave radiation. SWE measurements were obtained using a federal sampler (Osterhuber, 2014) at nine discrete sampling intervals during the March heatwave in the forest location.

2.3 Atmospheric structure and radiation

To investigate the synoptic patterns associated with the March heatwave, we use 500 hPa geopotential height and atmospheric column information from the ERA5 reanalysis (Hersbach et al., 2020). Daily geopotential height anomalies during the event are computed using data provided by Lee et al. (2023), which has been de-trended and coveres the period from 1979 to 2022.

We use the CERES-EBAF and CERES-SYN1Deg TODO: cite products to investigate the effects of clouds or lack-thereof on the snow energy balance. TODO: statement about EBAF. CERES-EBAF provides the gold-standard of measurement of surface and TOA irradiances at a 1x1 degree resolution at a monthly cadence. This product is suitable for trend-analysis. CERES-SYN1Deg is a lower level product that provides hourly estimates of the same quantities but is not configured to examine long-term temporal trends. Thus, we use CERES-SYN1Deg all-sky and clear-sky $SW\downarrow$ and $LW\downarrow$ (e.g., hypothetical irradiance in the absence of clouds) to compute cloud-radiative effects following Ramanathan et al. (1989) and subsequent studies. Rudisill et al. (2025) compared cloud radiative effects from CERES-SYN1Deg against in-situ radiometers in the Colorado Rockies and found good agreement, despite grid-resolution mismatches and confounding effects of terrain. The 2000-present period-of-record of CERES-EBAF are used to examine the March anomaly in $SW\downarrow$ and $LW\downarrow$ and the contribution of clouds to the anomalies therein.

2.4 SUMMA model configuration

To investigate the snow energy balance, we use the SUMMA model (Clark et al., 2015). SUMMA is well-vetted for a range of snow climate applications and provides several options for flux stability corrections, canopy radiation interactions, snow-layering schemes, and other model structural options (Cristea et al., 2022). We use the Jordan snow layering option, which allows for up to 100 snow layers, the UEB-2 stream approximation for canopy radiation interactions, the “Louisinv” option for stability correction of turbulent fluxes, and the “stickySnow” canopy snow interception scheme. SUMMA solves for a vegetation energy balance and canopy air temperature which is used to produce $LW\downarrow$ incident onto the sub-canopy snowpack. Additional specific modeling parameterizations with relevance to this study are described in Sec. 2.5 and the Supplemental Material. We assign a measurement height of 12m for the model forcings into the SUMMA in order to intercompare the effects of 10m tall needleleaf vegetation characteristic of the forest site at CSSL using the same forcing variables.

We use hourly data from the fifth-generation ECMWF (ERA5) reanalysis (Hersbach et al., 2020) as SUMMA model forcings after adjusting to the CSSL site. Daily precipitation from the NRCS SNOTEL site is used directly as forcing when available. Hourly ERA5 two-meter air temperatures are bias corrected to match the monthly average T_{air} observed at the CSSL from the NRCS records by removing a constant linear bias term. Comparing ERA5 $SW\downarrow$ data to observations from the CNR4 shows good agreement except for morning and evening when the surrounding vegetation blocks direct beam solar radiation. We compute and apply a $SW\downarrow$ correction coefficient as a function of the solar zenith angle to adjust the ERA5 data to account for shadows. To ensure that the ERA5 data matches the best available estimates of variability and trends in radiation at this site, we use a multiplicative scale that forces the ERA5 $SW\downarrow$ and $LW\downarrow$ monthly anomalies to match monthly anomalies in the CERES-EBAF data. Humidity, wind, and $LW\downarrow$ variables validated well against the available observations and are not adjusted. Additional information about SUMMA validation is provided in the supplemental material.

2.5 The snow energy balance

Using meteorological sign conventions for turbulent fluxes, the energy budget of a snowpack is given by:

$$\text{Melt/Refreeze} + \Delta\text{CC} = \underbrace{H_{\mathrm{s}} + H_{\mathrm{l}}}_{\text{Turbulent Fluxes}} + \underbrace{LW\downarrow - \sigma\epsilon T_{snow}^{4}}_{\text{longwave}} + \underbrace{SW\downarrow(1-\alpha)}_{\text{shortwave}} + G \quad (1)$$

,

where Melt/Refreeze is the rate of melt or liquid water refreezing in mass per second multiplied by the latent heat of fusion, ΔCC is the change in the snowpack cold-content, H_s and H_l are the turbulent fluxes of sensible and latent heat respectively at the snow/atmosphere interface, T_{air} is the temperature of the air,

T_{snow} is the temperature of the snow surface, ϵ is the emissivity of the snow (0.98), $SW\downarrow$ is the downwelling shortwave radiation, and α is the albedo of the snow surface defined as the ratio of upwelling shortwave radiation ($SW\uparrow$) to $SW\downarrow$.

The mass-balance of the snowpack is related to the energy balance (Eq. 1) in the following ways. The daily or monthly change in SWE is related to both Melt and to a lesser extent H_l, though the latter may remove 10% to upwards of 50% of the seasonal snow mass depending on a variety of factors (Sexstone et al., 2018; Schwat et al., 2025). Melt may be either positive (melt) or negative (refreezing) using the sign convention in Eq. 1. Melt integrated throughout an entire day is not synonymous with a loss in SWE, since meltwater may be retained in the snowpack and refreeze. In the SUMMA model, meltwater flow throughout the snowpack layers is modeled as a gravity-driven power law; preferential flow is not considered.

2.5.1 The influence of canopy Vegetation canopy influences the snow energy balance through several mechanisms. First, falling snow is intercepted by the canopy and may melt or sublimate from the canopy before reaching the snowpack, so total SWE accumulation under canopy is commonly less than adjacent open areas (CITE). Second, canopy layers may inhibit turbulent exchange of mass and energy by suppressing wind. Third and with most relevance to this work, vegetation impacts the radiative balance of the snowpack by altering $SW\downarrow$ and $LW\downarrow$ impacting the snow surface.

$$LW\downarrow_{\text{under canopy}} = \underbrace{(1-\varepsilon_c)\,LW\downarrow_{\text{atm}}}_{\text{sky LW through canopy gaps}} + \underbrace{\varepsilon_c\,\sigma\,T_c^4}_{\text{canopy emission}} \tag{2}$$

In SUMMA, the emissivity of the canopy layer (ϵ_c) is purely a function of the leaf-area index (LAI) and stem-area index.

2.5.2 Defining radiative forcing and effect We apply the radiative forcing concept in several ways in this study (Ramanathan et al., 1989; Deems et al., 2013; Rudisill et al., 2025). In this context, radiative forcing quantifies the instantaneous difference between the net or a single stream of radiation between observed conditions and a counterfactual reference state.

We define RF_{canopy} as the difference between R_{net} under the canopy relative to R_{net} outside of the canopy layer. Lacking observations sufficient to constrain RF_{canopy}, we use the SUMMA model to evaluate this quantity and express this value for both LW and SW components. To investigate the relative contribution of α decay on the surface radiation balance, we quantify the α radiative forcing (RF_{α}) by differencing SW_{net} computed using observed α versus SW_{net} computed using observed $SW\downarrow$ and a hypothetical fresh-snow broadband α; for this work we use 0.85. We define a cloud radiative "effect", rather than "forcing" to express the influence of clouds on downwelling $LW\downarrow$ and $SW\downarrow$ terms only (RE_{Cloud}). A full accounting of cloud radiative effects on net-energy terms also depends on the amount reflected and absorbed by the surface and thus the albedo (Rudisill et al., 2025) and assumptions about varying upwelling emissions. All radiative effects are signed such that positive indicates energy is added to the snowpack via that process (clouds, canopy, or α decay), and a negative sign indicates the opposite.

2.6 Ranking Snow Water Equivalent and Heatwave Anomalies

To rank or contextualize SWE anomalies, we use the framework from Hatchett et al. (2022), which uses a percentile-based metric to assign snow drought categories.

Table 1. Definitions of radiative effects used in this study. All effects are signed positive when energy is added to the snowpack. F refers the radiative component (SW or LW).

Name	Abbreviation	Equation
Cloud Radiative Forcing	RE_{cloud}	$\mathrm{F}_{\mathrm{cloudy}}\downarrow$ - $\mathrm{F}_{\mathrm{clear}}\downarrow$
Canopy Radiative Forcing	RF_{canopy}	$\mathrm{F}_{\mathrm{net,\ under\text{-}canopy}}$ - $\mathrm{F}_{\mathrm{net,\ above\text{-}canopy}}$
Albedo Radiative Forcing	RF_{α}	$SW\downarrow[(1\text{-}\alpha_{\mathrm{obs}})$ - $(1\text{-}\alpha_{\mathrm{clean}})]$

Drought categories are ranked as abnormally dry (D0), moderate drought (D1), severe drought (D2), extreme drought (D3), and exceptional drought (D4) for values between the 30th–20th, 20th–10th, 10th–5th, and 5th–2nd and below the 2nd percentiles, respectively. SWE percentiles are computed using the NRCS SNOTEL periods of record for each site.

Z-scores are used to rank near-surface T_{air} and snow ablation anomalies, given by

$$z = \frac{X - \mu}{\sigma}$$

. Baseline mean and standard deviations are computed using a common 1981-2020 normal periods unless otherwise stated.

3 Results

3.1 The March 2026 heatwave

The long-term and homogenized monthly USHCN HCNv2-data showed that 2026 had the highest March T_{air} observed since the beginning of the record in 1890, and was +3.6 °C above the 1981–2020 average — a full three-σ above the mean across the four-station composite (Fig. 2a). The nClimGrid likewise shows a +4–6 °C anomaly above average in some locations, compared to the +3.6°C anomaly observed at the four-station USHCN composite (Fig. 2b).

This prolonged period of heat resulted from an extreme and persistent ridge of high pressure that gained strength throughout March (Fig. 3a,b), with 500mb geopotential height anomalies exceeding 2-σ and 200m throughout March.

Immediately prior to the March 2026 heatwave, the region experienced three notable periods of weather. While the water-year-to-date precipitation (1 October—28 February) was 100% of 1991-2020 normal leading up to March 2026 (not shown), there was a prolonged midwinter dry spell (Hatchett et al., 2023) from 7 January to 9 February during which SWE began ablating (Fig. 3c). Then, between February 14th to 19th, a total of 100 mm of new SWE accumulated at CSSL, the third highest four-day total since records began in 1971. This snowfall event was followed by anomalously warm T_{air} and a rain-on-snow event on the 24—25 of February that brought 91 mm of rainfall, leading to a net decline in SWE (Fig. 3b,c).

The beginning of March started with average to slightly above average T_{air}. As the high pressure ridge grew over the WUS, daily maximum T_{air} peaked at 22.2°C on 19 March and 23.3°C on 20 March at CSSL but remained well-above normal from 8—31 March, with daily T_{air} values at the Tahoe City COOP station yielding z-scores greater than two-σ for four days between the 17th-27th. Correspondingly, the 500 mb height anomaly exceeded 200 meters across the WUS on this date, and was greater than two-σ above the detrended mean for this date. The hemispheric wave pattern on 16 March in particular is characteristic of an amplified North American Dipole pattern (Wang et al., 2014, 2015; Singh et al., 2016).

Snow began ablating consistently starting with the February ROS event for both the open and forest sites and continues into the month of March. We find good agreement between both SUMMA and observed SWE (Fig. 3c) and rates of ablation. SWE ablated at an average rate of 7.37 mm/day (7.06 mm/day) between the first of March and the 14th of March, after which it accelerated to a rate of 25.30 mm/day (18.66 mm/day) until the end of the month in the open (forest) until ablating completely in the open by the TODO: 28th and the forest site by the TODO: 24th.

3.2 SWE anomalies during water year 2026

The beginning of March saw SWE levels at a D1 or D0 ("No Drought") level across the twelve SNOTEL sites (Fig. 4a.) While SWE was nearly 0 mm for the two of the lowest elevation sites (station ID numbers 473 and 809) this was not particularly anomalous for those locations for this date. By the end of March, nine of the twelve stations reached 0 mm SWE and the remaining sites were in a D3 ("extreme drought") drought classification (Fig. 4b.). Only four of the SNOTEL sites had a precedent for 0 mm of SWE on March 31. Similarly, the snow covered area reached its lowest ever recorded value on March 31st value during the MODIS record (see supplemental material; Fig. S1). The rate of monthly snow ablation was likewise anomalous (Fig. 4c.) at most sites. Typically, most SNOTEL sites in this region gain (or remain constant) rather than lose SWE throughout March (Fig. 4b) with the exception of two of the lowest elevation sites. Nine out of twelve sites, including CSSL, measured their highest rate of monthly SWE ablation for the SNOTEL record. The lowest elevation site (473) is the only site without anomalous monthly SWE ablation, in part because the starting value was so close to zero.

3.3 Model and observed snowmelt rates and energy balance in the open and below canopy

To understand the causes of the anomalous observed ablation described in Sec. 3.2, we now focus on the CSSL site (site ID 428 in Fig. 4 and also depicted in Fig. 3c). The causes of the anomalous extreme ablation and the different rates between the forest and open sites can be understood by examining a time series of energy balance components (Eq. 1) and by separating them into nighttime, daytime, and 24-hour averages (Fig. 5a–f, Fig. 6a–f).

H_s and H_l are relatively minor factors compared to radiative fluxes (Fig. 5a,b, Fig. 6a,b). Two periods of significant sublimation occurred (negative H_l) on the 5th to the 7th, which correspond with relatively high wind speed (Fig. 3c) but prior to the onset of the warmest March temperatures. Another significant sublimation event took place on the 24th. The highest rates of H_s occurred between the 3rd through the 10th and again from the 21st to the 26th. A maximum daytime H_s exceeding 80 $\mathrm{W{\cdot}m^{-2}}$ took place on the 22nd for both the forest and open sites, just one day after the peak of the March temperatures.

SW_{net} terms far exceed the H_s maxima in the open site, however (Fig. 5c, Fig. 6c). Here, SW_{net} increases from daytime values approaching 80 $\mathrm{W{\cdot}m^{-2}}$ to values exceeding 240 $\mathrm{W{\cdot}m^{-2}}$ by the end of the month. Rates are significantly lower in the forest. SW_{net} increases throughout March, but never exceeds 80 $\mathrm{W{\cdot}m^{-2}}$.

In the forest, LW_{net} is positive on average for the entirety of March 2026 except the 4th-7th, which also corresponds with the only period of significant overnight cold-content production (Fig. 5d, Fig. 6d). Overnight LW_{net} switches from slight negative ($\approx$ -8 $\mathrm{W{\cdot}m^{-2}}$) to positive or near-zero by the 16th in the forest site. The highest rate of LW_{net} occurs on the 22nd, two days after the peak of the he. mat event, and exceeds 60 $\mathrm{W{\cdot}m^{-2}}$ during the daytime.

LW_{net} in the open site has a completely different sign, meaning that the snow is continuously losing heat via longwave emission, even during the peak of the heatwave in the latter half of March, and on the order of -80 to -40 W·m^{-2}.

The previous energy exchanges, in addition to ground heat flux (not shown), cumulatively manifest as energy directed towards breaking/reforming ice crystalline bonds (Melt/Refreeze) and warming ice to the melting point (ΔCC) A significant amount of energy is re-added back to the snowpack during the night by liquid water refreezing (6-33W·m^{-2}) in the open site, and to a lesser extent the forest site (Fig. 5e, Fig. 6e). The overnight LW_{net} cooling in the open yields overnight ΔCC that is eroded the following day, prior to the initiation of melt (yielding net-zero averages over a 24 hour period; Fig. 5f). The cumulative overnight production in cold-content becomes small compared to the following day's melt by the 14th of March and represents less than 20% of the total melt energy, indicating that the lack of overnight cold-content production — almost purely driven by the decline in LW_{net} and increase in overnight H_s— yields a snowpack that is primed for ablation the following day in the open site.

The dynamics of ΔCC are different in the forest— there is a near-zero amount of overnight cold-content production and very little refreezing. These model results are confirmed by field reports from CSSL, who found melf-refreeze crusts in the open during March but slush in the forest site even during morning observations, in agreement with these results.

3.4 Radiative Forcing

RF_{Albedo}, RF_{Canopy}, and RE_{Cloud} have a demonstrable impact on the energy balance, in that order (Fig. 7a—f). Snow α started at a relatively low value leading up to the March heatwave (0.6 to 0.7) and declines consistently during the month of March (Fig. 7a). Values between the observed albedo by the CSSL radiometers, SPIReS retrieval, and SUMMA generally agree, though SPIReS shows a lower value than either estimate. Both SPIReS and the CSSL albedo have several missing values towards the end of the event.

RF_{Albedo} increases at the open site from as a function of both rising sun-angle and $SW\downarrow$ (not shown) and α decline, reaching a value exceeding 60 W·m^{-2} in the open site based on the SUMMA simulation. While there is only a minor difference between the SUMMA model snow α between the forest and open sites, the RF_{Albedo} in the open exceeds RF_{Albedo} in the forest site by more than a factor of two by the end of the heatwave due to attenuation of $SW\downarrow$ by the canopy layer (Fig. 7b).

Cirrus clouds yield a modest by significant effect on the energy balance. RE_{Cloud} is negative for SW (decreasing incident sunlight) but positive for LW (increasing energy directed to the snowpack), with significant events happening on the first of the month, the 4th, the 10th, and the 21-23rd of the month.

Second in terms of magnitude to RF_{Albedo}, RF_{Canopy} yields a +20 to -90 W·m^{-2} maximum daily rate during the month (Fig. 7c). There are two competing effects. $\mathrm{RF}_{Canopy\ ,LW}$ increases as a function of T_{air} at a rate of 3.3 W·m^{-2} per °C of T_{air}, which is visualized by considering the daily relationship for the entire SUMMA output record for the month of March. Additional analysis shows that the canopy layer is consistently warmer than the ambient T_{air}, which combined with the emissivity relationship (Eq. 2) explains this effect. This effect does not exceed the shading effect of the canopy ($\mathrm{RF}_{Canopy\ ,SW}$) however, even though the canopy attenuates much of the incident sunlight, so $\mathrm{RF}_{Canopy\ ,Net}$ is near-zero and transitions to negative (cooling the snow) during the hottest part of the heatwave.

Ultimately, the cumulative energy transmitted from the atmosphere to the snow attributed to RF_{Albedo} is 101.3 and 23.7 $\mathrm{MJ/m^2}$ for the open and forest site respectively and RF_{Canopy} is responsible for a -95.1 $\mathrm{MJ/m^2}$ effect (reducing energy exchange). The definition of RE_{Cloud} is simply the difference between clear and all-sky downwelling terms, and clouds reduce $SW\downarrow$ (-49 $\mathrm{MJ{\cdot}m^{-2}}$) and increase $LW\downarrow$ (+31 $\mathrm{MJ{\cdot}m^{-2}}$). A full accounting of the net-effect on the snow energy balance also requires taking into account what is reflected and absorbed by the snow. Assuming a weighted albedo value of 0.6 (and that the CERES SYN1Deg data is applicable at this scale) a coarse estimate of the net-cloud radiative forcing yields 11 $\mathrm{MJ/m^2}$ of heating to the snow in the open (Fig. 7f). To place these value in context, comparisons against climatological RE_{Cloud} are presented in Section 3.5.

We do not compute a similar value of for the forest site since the co-variance with the vegetation energy and radiative transfer does not yield a pragmatic solution, but we hypothesize that the magnitude may be similar due to the competing effects from SW and LW mechanisms.

3.5 Climatological context of the March 2026 snow energy balance

How does the March 2026 energy balance compare to typical Marches at CSSL? The daily components of the energy budget can be contextualized by examining cumulative sums, in $\mathrm{MJ{\cdot}m^{-2}}$, of the different energy balance terms to quantify their total contributions to ablation (Fig. 8a–g). The accumulated SW_{net} is both the strongest forcing (+223.6 $\mathrm{MJ{\cdot}m^{-2}}$) and also the most anomalous forcing compared to the 1980-present March average (+72.0 $\mathrm{MJ{\cdot}m^{-2}}$), followed by H_s (+45.6 $\mathrm{MJ{\cdot}m^{-2}}$, with a +18.7 anomaly) in the open site. For the forest site, SW_{net} is also the strongest forcing (+48.3 $\mathrm{MJ{\cdot}m^{-2}}$) but not the most anomalous. This designation applies to LW_{net}, which is the second strongest forcing (+27.8 $\mathrm{MJ{\cdot}m^{-2}}$) with an anomaly of +28.9 $\mathrm{MJ{\cdot}m^{-2}}$ relative to historical conditions, indicating that — for a typical year — LW_{net} is below-zero (and a source of snow cooling) rather than providing energy for heating snow under canopy.

The above-canopy $SW\downarrow$ and $LW\downarrow$ were particularly anomalous during March 2026 (Fig. 8a). To understand these anomalies, we examine map-views and timeseries of the CERES EBAF product for the WUS and northern Sierra Nevada subset region (Fig. 9a–c). March of 2026 had the highest $SW\downarrow$ to the entire CERES record dating back to March of 2000 and is more than 1.5 standard deviations above average for the subset region. Nearly 120 $\mathrm{MJ{\cdot}m^{-2}}$ of additional $SW\downarrow$ energy irradiated the land surface compared to average conditions. While less anomalous in terms of z-score, the $LW\downarrow$ anomaly is over 24 $\mathrm{MJ{\cdot}m^{-2}}$ above average and the second-highest rate of monthly $LW\downarrow$ during the CERES EBAF record for this region.

Variations in the monthly $SW\downarrow$ are almost entirely driven by variations in cloud radiative effect, though the same is not true for $LW\downarrow$ (Fig. 9c,d). RE_{Cloud} values were the highest (i.e., the weakest rate of cooling) recorded by CERES EBAF, and correspondingly the RE_{Cloud} LW effect was also the lowest on record, which together indicate that skies were anomalously free of optically thick clouds during this period. Correspondingly, no precipitation was observed during the month (Fig. 3c). With this in mind, it is even more remarkable that $LW\downarrow$ reached the second-highest observed rate for this region despite a clear-atmosphere, since clouds radiative as a nearly perfect gray-body at their base-temperature in contrast with the clear-sky atmosphere (Shupe and Intrieri, 2004).

4 Discussion

5 Conclusions

In this study, we used energy balance observations from CSSL, NRCS SNOTEL observations, snow energy balance modeling (SUMMA), atmospheric reanalyses (ERA5) and remote sensing products (CERES EBAF and SYN1Deg) to investigate interactions between anomalously warm air temperatures and snow ablation during the month of March 2026. The unprecedented rates of ablation were driven by several factors. The 3-σ March T_{air} anomaly also corresponded with the highest rate of $SW\downarrow$ observed during the CERES record (2000 to present) across the study region, driven by a near-complete lack of mid or low-level stratiform clouds. Combined with the complete absence of new snowfall, snow albedo declined rapidly.

The higher-than average $SW\downarrow$ combined with well-below average snow albedo meant that net-shortwave rapidly ablated the snow in the later half of March. Vegetation emits longwave radiation at higher rates as the temperatures rose, but the added longwave to the sub-canopy snowpack was outweighed by shading effects, meaning that both snow ablates at a lower rate under canopy relative (XX mm/day) relative to the open site (XX mm/day). Overnight cold-content production influences variability in snow ablation. But, the extreme monthly rates of ablation are insensitive to reasonable variations in the initial thermodynamic state of the snowpack.

Sensible heating and sublimation were relatively minor factors, and the latter contributed less than 1% of the total snow ablation. Theoretical arguments show that, over 0°C snow, that sublimation is permitted by an increasingly narrow range of atmospheric humidity values were not consistently maintained during the month of March. Likewise, atmospheric buoyancy effects mitigate sensible heating at high air temperatures and low winds. While most models make these assumptions, more work is needed to validate turbulent exchanges into snowpacks in complex terrain and in the vicinity of vegetation canopy where these assumptions may not hold. While snow-albedo parameterizations performed well against CSSL data, we propose that future observations should focus on albedo decay processes and grain-growth mechanisms during extreme heatwave events.

Heatwave events such March 2026 are projected to increase with climate warming, necessitating efforts to better model and adapt water resource infrastructure and investigate ecosystem management for resiliency. Synoptic analyses of snow-eater heatwave events should focus on the co-variation of heatwave conditions with radiative anomalies and the dynamics that lead to the prolonged cloud-free conditions that contributed to the anomalous radiation conditions observed during March 2026.

6 Data Availability

ERA5 atmospheric data and near-surface meteorological forcings for SUMMA were downloaded from the Copernicus data store ECMWF (2023). Geopotential height anomalies and standard deviations were precomputed and provided from the supplemental material of Lee et al. (2023). SPIReS near real time data are publicly available from the National Snow and Ice Data center Snow Today page. The SUMMA model is publicly available on GitHub. The VIIRS/Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature was processed and downloaded from Google Earth Engine TODO: CITE GEE.

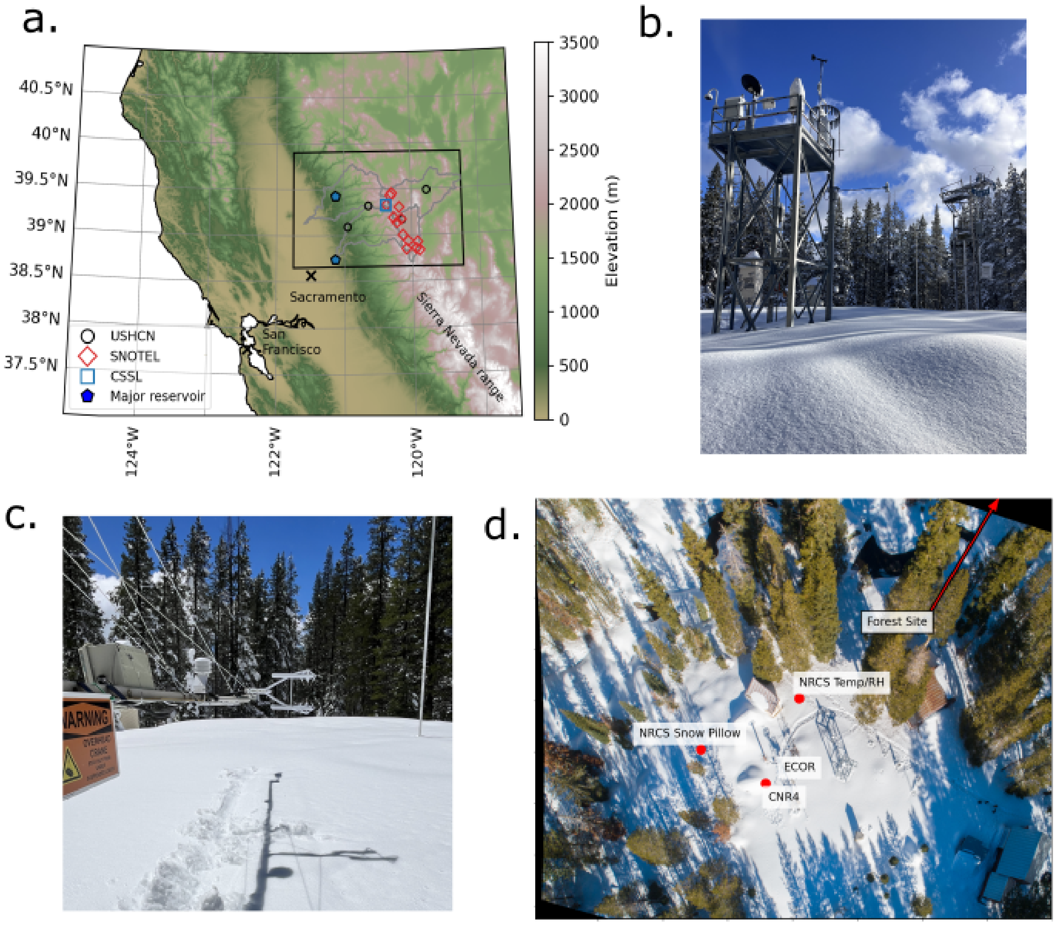

Figure 1. TODO: change the font so it's the same as the other figures**(a)** The northern California Sierra Nevada study region with terrain elevation. Outlines depict the Yuba, North Fork American, Lake Tahoe, and Truckee HUC8 watershed boundaries. Red open diamonds depict the location of NRCS SNOTEL sites, open black circles show the location of the four USHCN stations in the region, and the open blue square depicts the CSSL energy balance site. **(b)** The CSSL meteorological tower in winter, **(c)** a close-up view of the Campbell CNR4 radiometer and IRGASON eddy covariance system, and **(d)** an aerial overview of the CSSL site. The location of the forest meteorological tower (not shown) is located just outside of the frame.

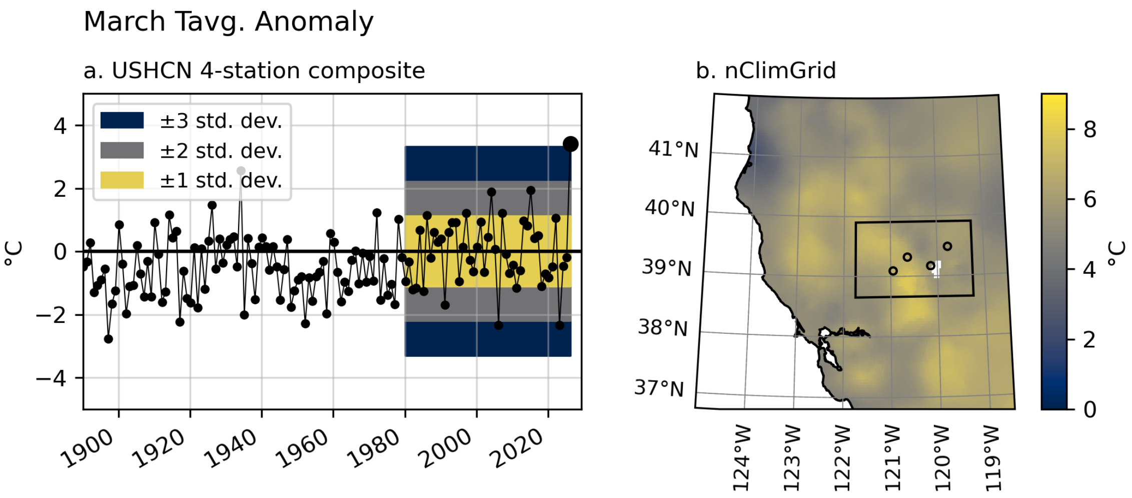

Figure 2. (a) Composite T_{air} anomalies from the four HCNv2 dataset stations in the greater Tahoe region study area for the month of March, from 1890 to 2026. Deviations from the 1980–2026 mean are depicted for one, two, and three standard deviation regions (shading). Anomalies are computed with respect to the 1980–2026 mean T_{air}. (b) The same, but for a map-view of the T_{air} anomaly of the gridded nClimGrid product.

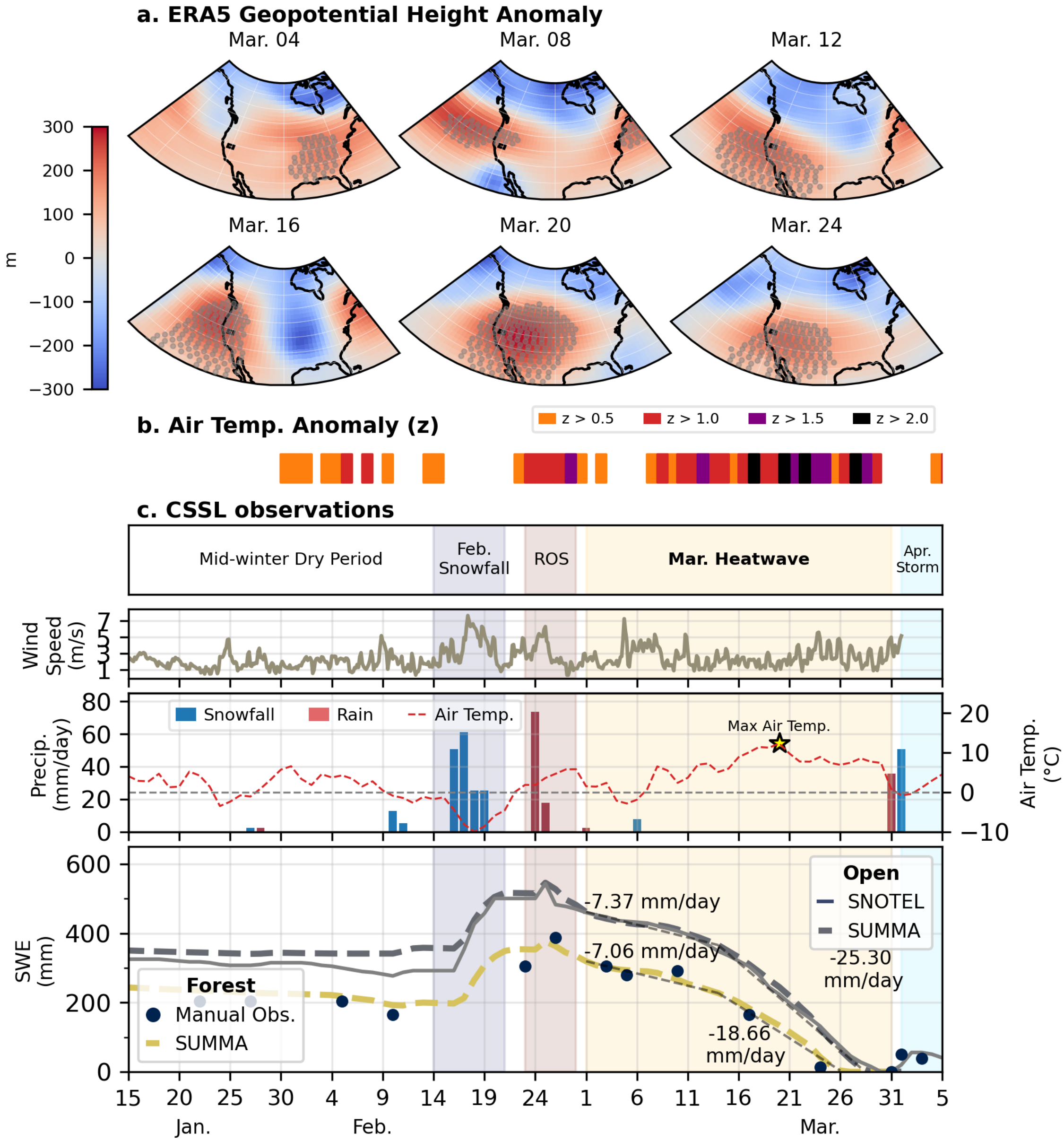

Figure 3. TODO: Change "Melt" to "Ablation". Change mar to March and Feb to February The meteorological conditions leading up to and during the March 2026 heatwave. **(a)** The 500 mb geopotential height anomaly during the March 2026 heatwave from ERA5 for six days across the WUS. The study region is outlined using an overlain black rectangle. Gray stippling indicates regions with 500 mb geopotential height anomalies greater than 2-σ above the 1950-2022 mean for the month of March. **(b-c)** shows the time series of meteorological conditions at CSSL before and during the heatwave event. **(b)** depicts the daily average two-meter air temperature and daily precipitation rate. Days with a majority of rain (red) and a majority of snowfall (blue) are differentiated. **(c)** depicts the daily SWE and the one-day change in SWE (dSWE; right axis). Days with a positive dSWE (blue) are denoted by accumulation, and days with a negative dSWE (red) are termed melt days, acknowledging that some SWE may be lost by other processes.

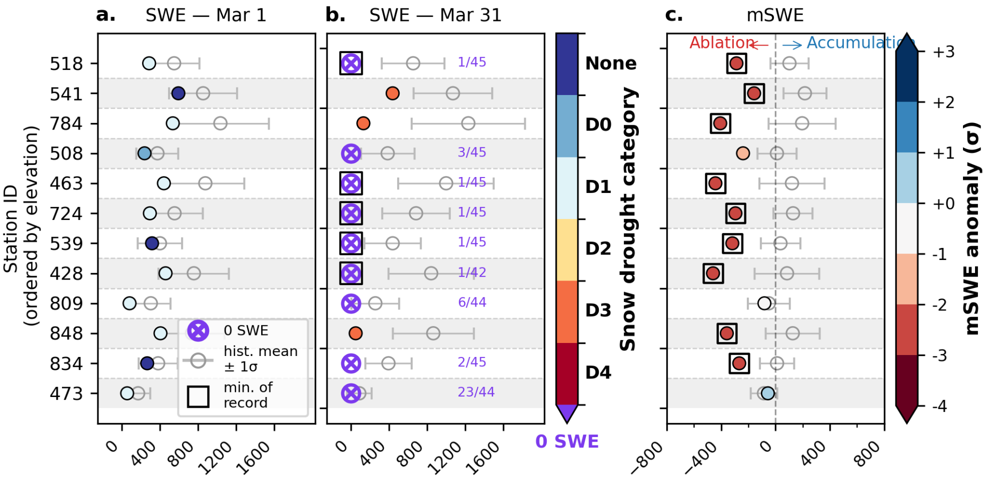

Figure 4. (a) SWE values on March 1st the 12 SNOTEL sites in the study domain, labeled by their three-digit station code. (b) The same as (a), but for SWE on March 31. Color values correspond with the snow drought D-scale. 0 SWE values are depicted by purple circle markers and the purple text lists how common 0 SWE was observed on that date for that site out of the period of record. (c) The SWE change between Mar 31 and Mar 1 compared with the historical record. Color values depict the deviation from the mean in units of standard deviation. Square markers bounding symbols indicates the SWE or mSWE is the record minimum for the period of record.

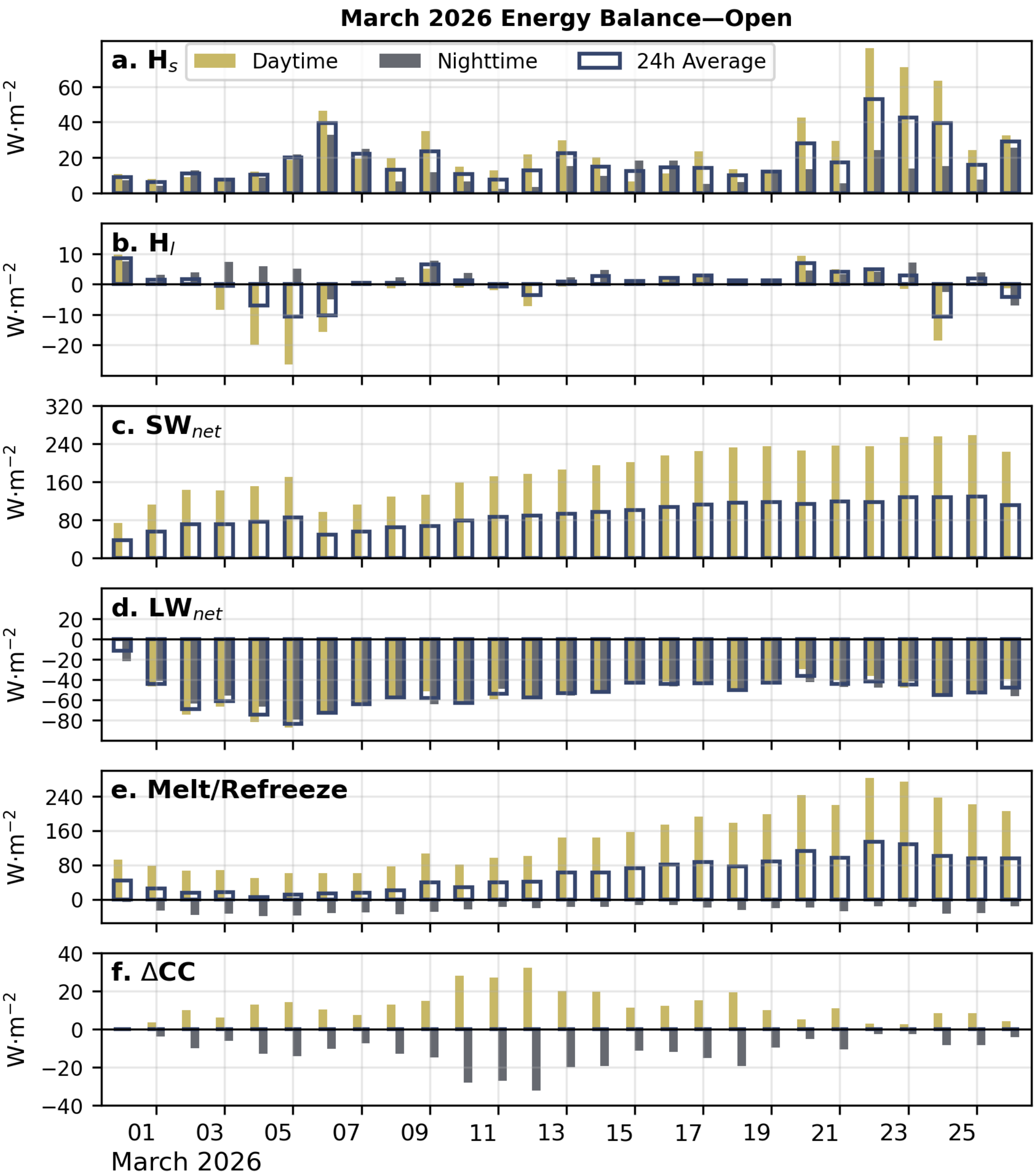

Figure 5. Daily average energy flux components including (a) H_s, (b) H_l, (c) SW_{net}, (d) LW_{net}, (e) snowpack Melt/Refreeze and (f) change in cold-content (Δ CC) during the March 2026 heatwave the open site. Each day has three bars: the left-most yellow bar depicts the daytime flux, the right-most gray bar depicts the nighttime flux, and the outline spanning both bars depicts the 24-hour average flux. Please note that the y-scales are not equivalent across all of the figures.

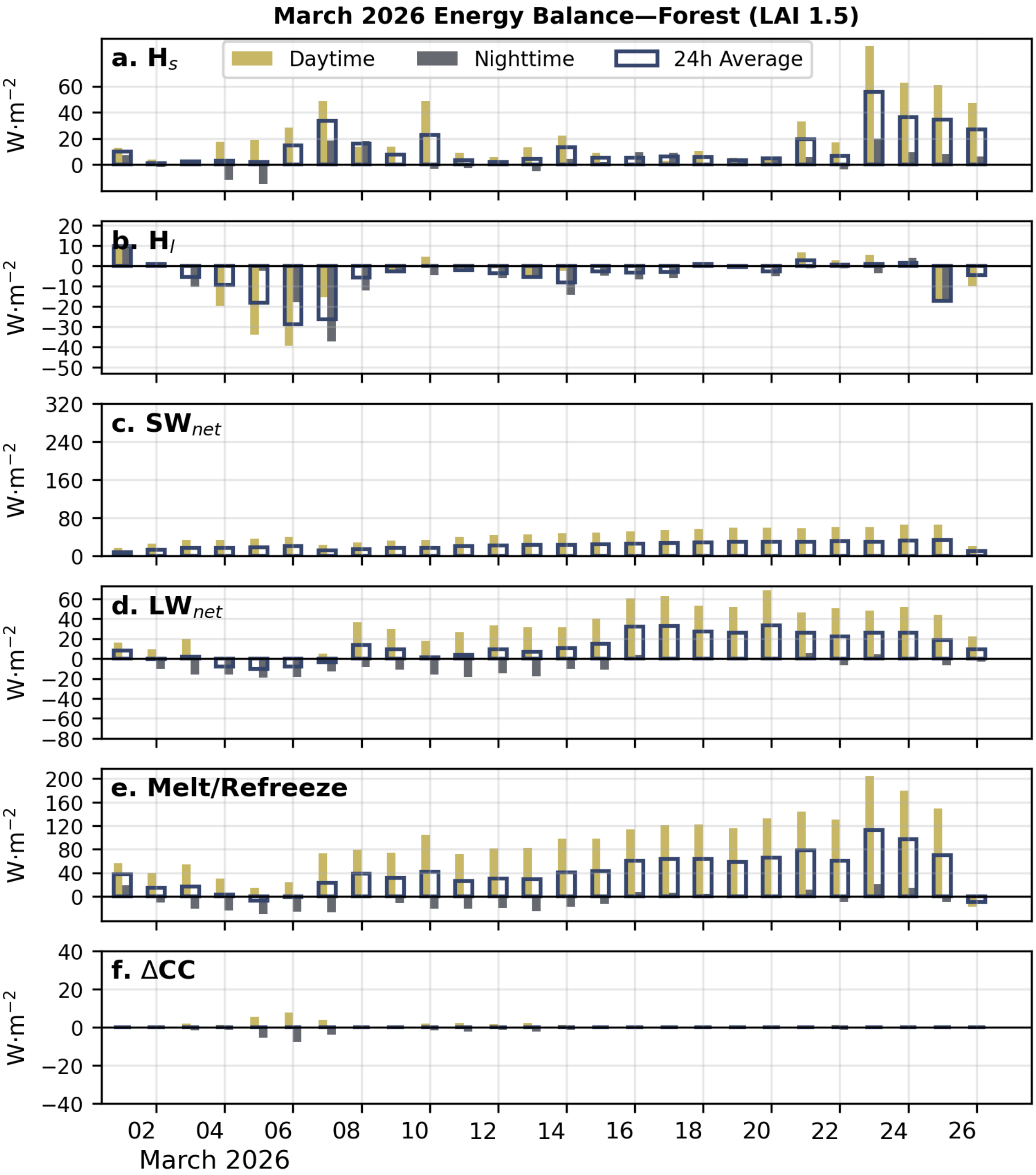

Figure 6. The same as Fig. 5 but for the forest site. Note the y-axis limits and ticks are not the same as (Fig. 5)

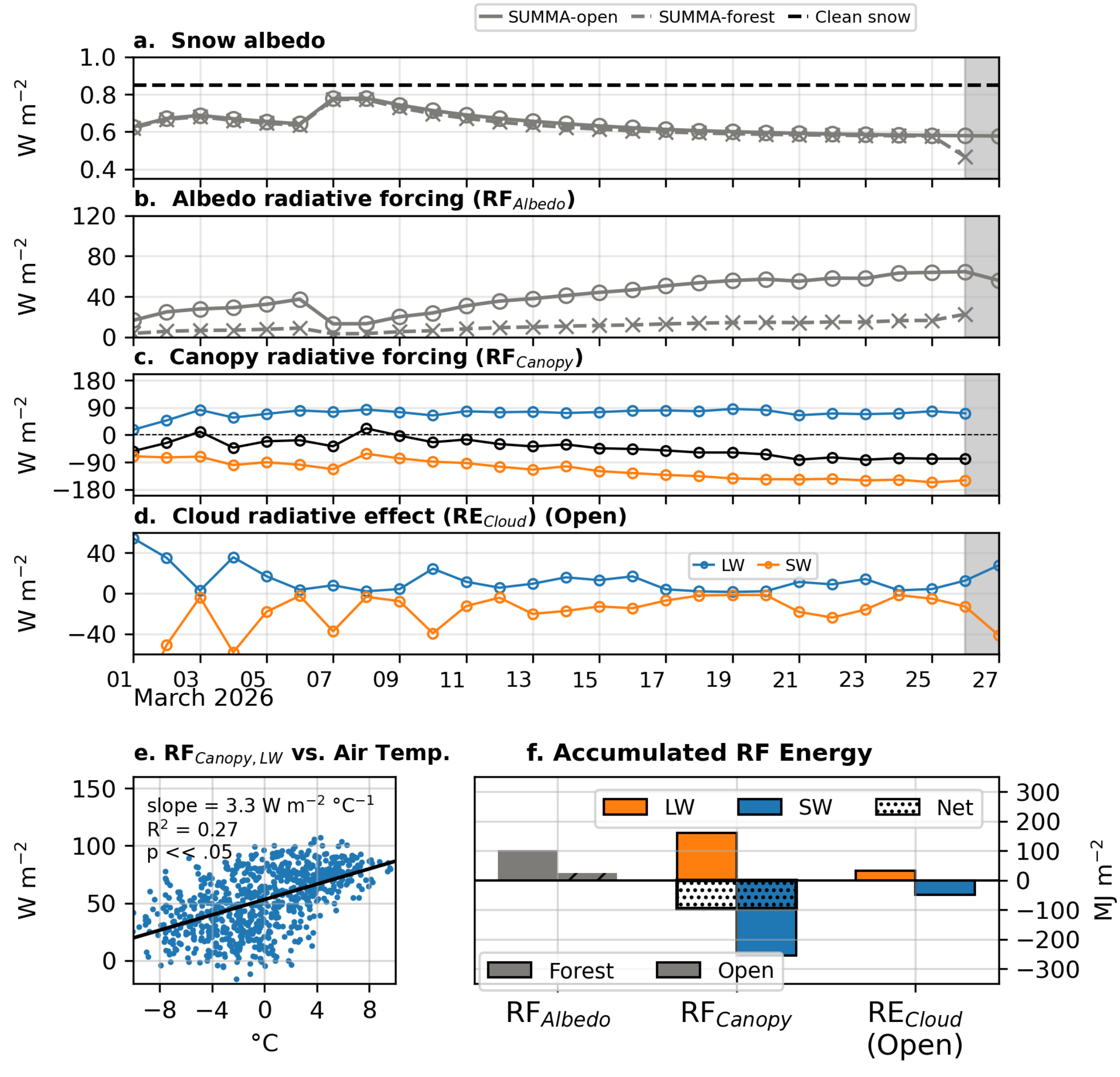

Figure 7. test

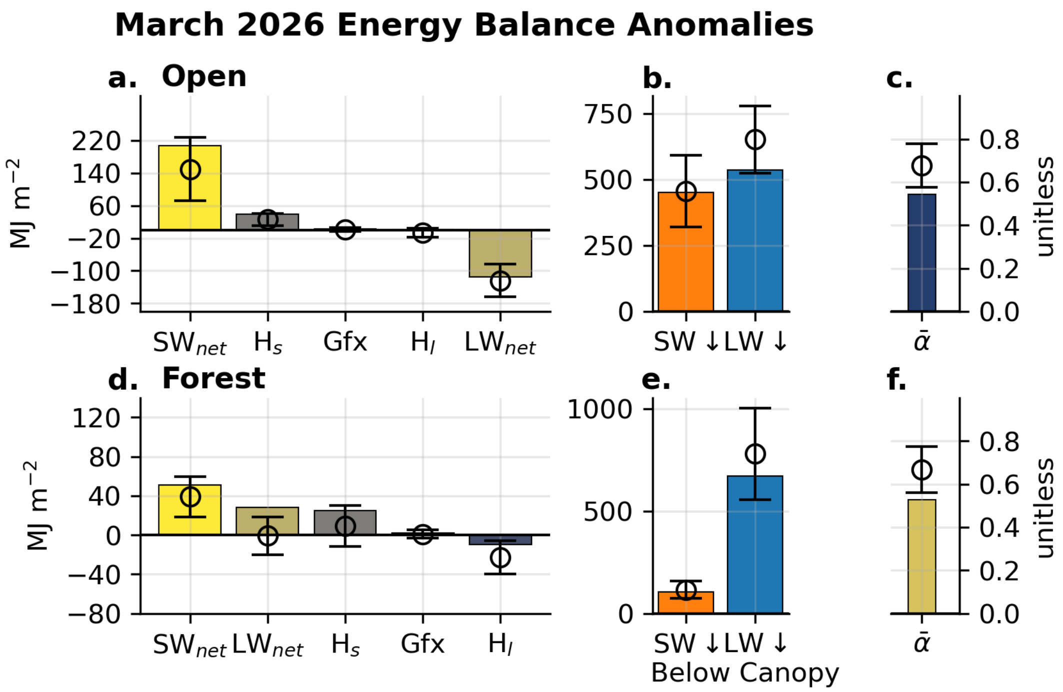

Figure 8. March-accumulated surface energy fluxes, snowmelt/cold-content responses, and snow albedo for the open (a–c) and forest (d–f) sites, restricted to hours with SWE > 4 mm. Solid bars gives the accumulated (or, for albedo, the incoming-shortwave-weighted mean) value for March 2026, and the adjacent boxplot (outliers omitted) shows the distribution of the same quantity across all March 1980–2025 water years. (a,d) Monthly-accumulated net shortwave (SW$_{net}$), net longwave (LW$_{net}$), sensible (H$_s$) and latent (H$_l$) heat, and ground heat flux (G) [MJ m^{-2}] for the open (a) and forest (d) sites; The inset in (a) shows the analogous accumulated $SW\downarrow$ and $LW\downarrow$ radiative forcing. (b,e) Accumulated total snowmelt energy (Melt) and change in snowpack cold content (dCC) for the open (b) and forest (e) sites. (c,f) Shortwave-weighted mean snow albedo (α) for the open (c) and forest (f) sites, computed by weighting each hour's albedo by its share of that month's total incoming shortwave radiation.

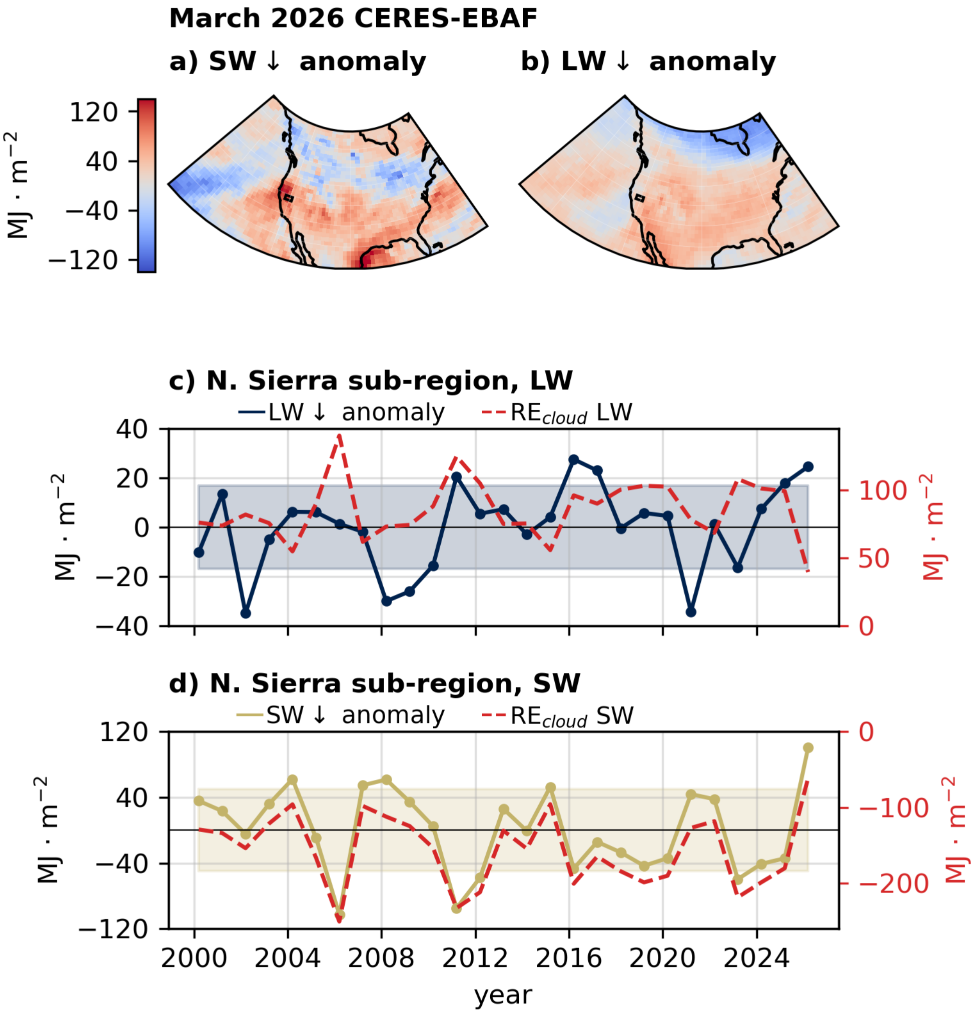

Figure 9. March $SW\downarrow$ (a) and $LW\downarrow$ (b) anomalies (relative to the March 2000 to present record) from the CERES EBAF 1x1 degree resolution product. (c) and (d) depicts the timeseries of anomalies averaged over the northern Sierra Nevada region in addition to cloud radiative effect amount (red), with units displayed on the right y-axis (in red-text). The shaded region in each plot depicts one standard deviation of the anomaly.

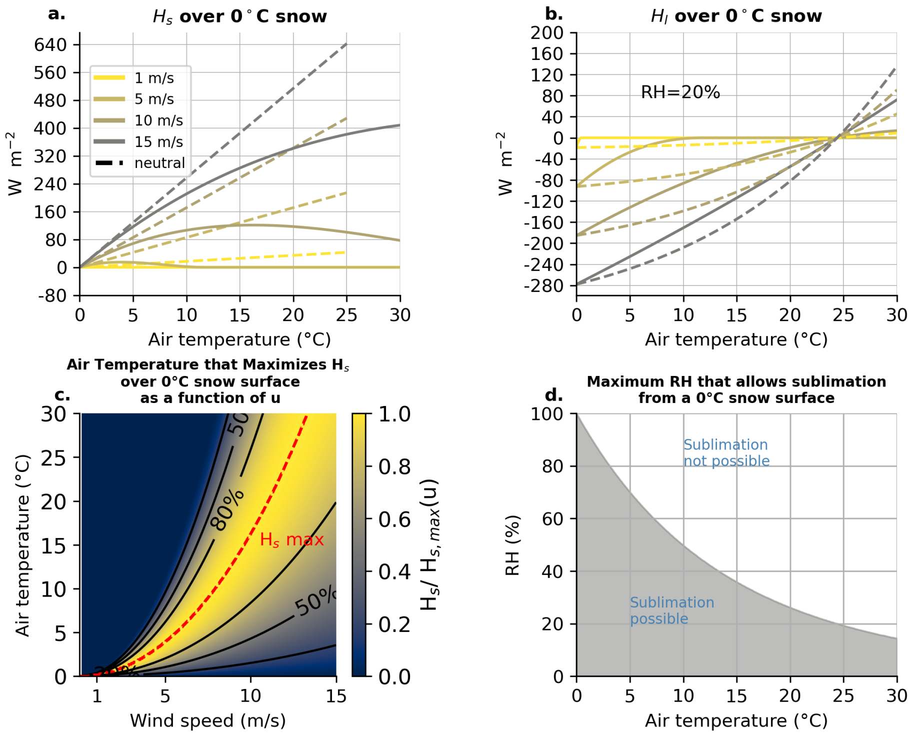

Figure 10. (a) Idealized sensible heat fluxes (H_s) into a melting snowpack as a function of T_{air} for four wind speeds (1, 5, 10, and 15 m s^{-1}) at a 12 m height for open-conditions and a snow surface roughness of z$_0$=0.002 m. Dashed lines show neutral-stability bulk transfer; solid lines show the same formulation with a stability correction applied as implemented in SUMMA. (b) the same as (a), but for H_l with a fixed RH of 25%. (c) The T_{air} that yields the maximum H_s into a 0 °C snowpack after correcting for stability effects (dashed red line) for a given wind speed (u). Contours depict the fraction of the maximum H_s for that wind speed value. (d) The thermodynamic limitation of sublimation as a function of T_{air}, depicting the maximum RH that permits sublimation from a 0 °C snowpack.

	SW_{net} (Open)	SW_{net} (Forest)	LW_{net} (Open)	LW_{net} (Forest)	H_l (Open)
2026	223.55	48.31	-127.49	27.83	0.06
mean	151.53	39.25	-126.27	-1.09	-7.34
+/−2-std	67.82	17.54	42.11	17.52	10.54

References

E. A. Anderson. *National Weather Service River Forecast System: Snow Accumulation and Ablation Model*. U.S. Department of Commerce, National Oceanic and Atmospheric Administration, National Weather Service, 1973.

E. Bercos-Hickey, T. A. O'Brien, M. F. Wehner, L. Zhang, C. M. Patricola, H. Huang, and M. D. Risser. Anthropogenic contributions to the 2021 pacific northwest heatwave. *Geophysical Research Letters*, 49(23):e2022GL099396, 2022. doi: https://doi.org/10.1029/2022GL099396. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2022GL099396. e2022GL099396 2022GL099396.

M. P. Clark, B. Nijssen, J. D. Lundquist, D. Kavetski, et al. A unified approach for process-based hydrologic modeling: 1. modeling concept. *Water Resour. Res.*, 51 (4):2498–2514, 2015. doi: 10.1002/2015wr017198.

M. Cowherd, U. Mital, S. Rahimi, M. Girotto, A. Schwartz, and D. Feldman. Climate change-resilient snowpack estimation in the western united states. *Commun. Earth Environ.*, 5(1):337, 2024. doi: 10.1038/s43247-024-01496-3.

N. C. Cristea, A. Bennett, B. Nijssen, and J. D. Lundquist. When and where are multiple snow layers important for simulations of snow accumulation and melt? *Water Resour. Res.*, 58(10):e2020WR028993, 2022. doi: 10.1029/2020wr028993.

J. S. Deems, T. H. Painter, J. J. Barsugli, J. Belnap, and B. Udall. Combined impacts of current and future dust deposition and regional warming on colorado river basin snow dynamics and hydrology. *Hydrol. Earth Syst. Sci.*, 17(11): 4401–4413, Nov. 2013. ISSN 1027-5606,1607-7938. doi: 10.5194/hess-17-4401-2013.

ECMWF. CDS API – Copernicus climate data store python interface. https://github.com/ecmwf/cdsapi, 2023. Accessed: 2026-05-06.

K. E. Hale, K. S. Jennings, K. N. Musselman, B. Livneh, and N. P. Molotch. Recent decreases in snow water storage in western north america. *Communications Earth & Environment*, 4(1):1–11, May 2023. ISSN 2662-4435. doi: 10.1038/s43247-023-00751-3.

B. J. Hatchett, A. M. Rhoades, and D. J. McEvoy. Monitoring the daily evolution and extent of snow drought. *Natural Hazards and Earth System Sciences*, 22(3): 869–890, 2022. doi: 10.5194/nhess-22-869-2022. URL https://nhess.copernicus.org/articles/22/869/2022/.

B. J. Hatchett, A. L. Koshkin, K. Guirguis, K. Rittger, A. W. Nolin, A. Heggli, A. M. Rhoades, A. E. East, E. R. Siirila-Woodburn, W. T. Brandt, A. Gershunov, and K. Haleakala. Midwinter dry spells amplify post-fire snowpack decline. *Geophysical Research Letters*, 50(3):e2022GL101235, 2023. doi: https://doi.org/10.1029/2022GL101235. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2022GL101235. e2022GL101235 2022GL101235.

A. Heggli, B. Hatchett, A. Schwartz, T. Bardsley, and E. Hand. Toward snowpack runoff decision support. *iScience*, 25(5), 2022. doi: 10.1016/j.isci.2022.104240. URL https://doi.org/10.1016/j.isci.2022.104240.

A. Heggli, B. J. Hatchett, A. Chen, T. Bardsley, J. Moore, M. Imgarten, and P. Fickenscher. The application of a snowpack runoff decision support system for rain-on-snow events. *Water Resources Research*, 62(4):e2025WR042453, 2026. doi: https://doi.org/10.1029/2025WR042453. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2025WR042453. e2025WR042453 2025WR042453.

H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. De Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, and J.-N. Thépaut. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730): 1999–2049, 2020. doi: 10.1002/qj.3803.

R. S. Kim, S. Kumar, C. Vuyovich, P. Houser, J. Lundquist, L. Mudryk, M. Durand, A. Barros, E. J. Kim, B. A. Forman, E. D. Gutmann, M. L. Wrzesien, C. Garnaud, M. Sandells, H.-P. Marshall, N. Cristea, J. M. Pflug, J. Johnston, Y. Cao, D. Mocko, and S. Wang. Snow ensemble uncertainty project (SEUP): quantification of snow water equivalent uncertainty across north america via ensemble land surface modeling. *The Cryosphere*, 15(2):771–791, 2021. doi: 10.5194/tc-15-771-2021.

T. S. L'Ecuyer, Y. Hang, A. V. Matus, and Z. Wang. Reassessing the effect of cloud type on earth's energy balance in the age of active spaceborne observations. part I: Top of atmosphere and surface. *J. Clim.*, 32(19):6197–6217, 2019. doi: 10.1175/jcli-d-18-0753.1.

S. H. Lee, M. K. Tippett, and L. M. Polvani. A new year-round weather regime classification for north america. *J. Clim.*, 36(20):7091–7108, Sept. 2023. ISSN 0894-8755,1520-0442. doi: 10.1175/JCLI-D-23-0214.1.

D. Li, M. L. Wrzesien, M. Durand, J. Adam, and D. P. Lettenmaier. How much runoff originates as snow in the western united states, and how will that change in the future? *Geophys. Res. Lett.*, 44(12):6163–6172, June 2017. ISSN 0094-8276,1944-8007. doi: 10.1002/2017gl073551.

S. A. McAfee, G. J. McCabe, S. T. Gray, and G. T. Pederson. Changing station coverage impacts temperature trends in the upper colorado river basin. *Int. J. Climatol.*, 39(3):1517–1538, Mar. 2019. ISSN 0899-8418. doi: 10.1002/joc.5898.

M. J. Menne, C. N. Williams, and R. S. Vose. The U.S. historical climatology network monthly temperature data, version 2. *Bulletin of the American Meteorological Society*, 90(7):993–1007, 2009. doi: 10.1175/2008BAMS2613.1.

M. J. Menne, I. Durre, R. S. Vose, B. E. Gleason, and T. G. Houston. An overview of the global historical climatology network-daily database. *J. Atmos. Ocean. Technol.*, 29(7):897–910, July 2012. ISSN 0739-0572,1520-0426. doi: 10.1175/jtech-d-11-00103.1.

R. Osterhuber. Snow survey procedure manual. *The Resources Agency of California*, 2014.

J. W. Oyler, S. Z. Dobrowski, A. P. Ballantyne, A. E. Klene, and S. W. Running. Artificial amplification of warming trends across the mountains of the western united states. *Geophys. Res. Lett.*, 42(1):153–161, 2015. ISSN 0094-8276. doi: 10.1002/2014gl062803.

V. Ramanathan, R. D. Cess, E. F. Harrison, P. Minnis, B. R. Barkstrom, E. Ahmad, and D. Hartmann. Cloud-radiative forcing and climate: results from the earth radiation budget experiment. *Science*, 243(4887):57–63, Jan. 1989. ISSN 0036-8075. doi: 10.1126/science.243.4887.57.

A. M. Rhoades, J. S. North, W. Rudisill, B. J. Hatchett, M. Risser, A. Beltran-Peña, A. Heggli, S. Hotaling, L. S. Huning, A. Joros, M. LaPlante, A. Mahesh, A. M. Marshall, R. McCrary, D. McEvoy, S. Rahimi, M. S. Raleigh, C. Randall, A. Srivastava, M. Wehner, Y. Zhou, and A. D. Jones. Snow-eater heat waves of the western united states. *Sci. Adv.*, 12(32):eaeb3361, 2026. doi: 10.1126/sciadv.aeb3361.

W. Rudisill, D. Feldman, C. J. Cox, L. Riihimaki, et al. Seasonality and albedo dependence of cloud radiative forcing in the upper colorado river basin. *J. Geophys. Res.*, 130(6):e2024JD042366, 2025. doi: 10.1029/2024JD042366.

A. Schwartz, M. Cowherd, B. McGurk, M. Mason, K. Riker-Coleman, B. Zutter, and G. Lewis. Modernization of the central sierra snow laboratory – a new chapter in a storied existence. *Bulletin of the American Meteorological Society*, Jun 2026. doi: 10.1175/bams-d-25-0112.1.

E. Schwat, D. Hogan, K. T. Paw U, C. J. Cox, et al. Estimating snow sublimation in complex terrain: A season of intensive field measurements and the role of vertical water vapor flux divergence. *J. Hydrometeorol.*, 26(10):1455–1473, 2025. doi: 10.1175/jhm-d-25-0022.1.

M. C. Serreze, M. P. Clark, R. L. Armstrong, D. A. McGinnis, and R. S. Pulwarty. Characteristics of the western united states snowpack from snowpack telemetry (SNOTEL) data. *Water Resour. Res.*, 35(7):2145–2160, 1999. ISSN 0043-1397. doi: 10.1029/1999wr900090.

G. A. Sexstone, D. W. Clow, S. R. Fassnacht, G. E. Liston, C. A. Hiemstra, J. F. Knowles, and C. A. Penn. Snow sublimation in mountain environments and its sensitivity to forest disturbance and climate warming. *Water Resour. Res.*, 54(2): 1191–1211, 2018. doi: 10.1002/2017wr021172.

M. D. Shupe and J. M. Intrieri. Cloud radiative forcing of the arctic surface: The influence of cloud properties, surface albedo, and solar zenith angle. *J. Clim.*, 17 (3):616–628, Feb. 2004. ISSN 0894-8755. doi: 10.1175/1520-0442(2004)017⟨0616:CRFOTA⟩2.0.CO;2.

D. Singh, D. L. Swain, J. S. Mankin, D. E. Horton, L. N. Thomas, B. Rajaratnam, and N. S. Diffenbaugh. Recent amplification of the north american winter temperature dipole. *J. Geophys. Res Atmos.*, 121(17):9911–9928, 2016. doi: 10.1002/2016JD025116.

S.-Y. Wang, L. Hipps, R. R. Gillies, and J.-H. Yoon. Probable causes of the abnormal ridge accompanying the 2013–2014 california drought: Enso precursor and anthropogenic warming footprint. *Geophysical Research Letters*, 41(9): 3220–3226, 2014. doi: https://doi.org/10.1002/2014GL059748. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2014GL059748.

S.-Y. S. Wang, W.-R. Huang, and J.-H. Yoon. The north american winter 'dipole' and extremes activity: a cmip5 assessment. *Atmospheric Science Letters*, 16(3): 338–345, 2015. doi: https://doi.org/10.1002/asl2.565. URL https://rmets.onlinelibrary.wiley.com/doi/abs/10.1002/asl2.565.